

Boss Challenge — Paper 38 (Class 7)

Winds, Storms & Cyclones | Wastewater Story | Exponents & Powers | Lines & Angles

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Air around us pushes on every surface it touches; this push spread over an area is called air:
(A) humidity
(B) vapour
(C) pressure
(D) speed
2. In the number 2^5 , the small raised 5 is called the:
(A) exponent
(B) base
(C) product
(D) sum
3. The used, dirty water released from homes, factories and farms is called:
(A) wastewater
(B) spring water
(C) rainwater
(D) distilled water
4. When the measures of two angles total exactly 90° , the pair is described as:
(A) complementary angles
(B) vertically opposite angles
(C) reflex angles
(D) supplementary angles
5. When air is heated it expands, becomes lighter, and tends to:
(A) sink down
(B) turn solid
(C) rise up
(D) stop moving
6. In 7^3 , the number 7 that is multiplied repeatedly is called the:
(A) base
(B) exponent
(C) product
(D) remainder
7. Wastewater that carries human waste and dirt away from homes through pipes is also called:
(A) nectar
(B) groundwater
(C) sewage
(D) dew

8. When the measures of two angles total exactly 180° , the pair is described as:
- (A) acute angles
 - (B) complementary angles
 - (C) corresponding angles
 - (D) supplementary angles
9. Wind is simply air that is:
- (A) trapped in a sealed jar
 - (B) completely still
 - (C) frozen into ice
 - (D) moving from one place to another
10. The repeated product $2 \times 2 \times 2 \times 2$ can be written in short as:
- (A) 2×4
 - (B) 8^2
 - (C) 4^2
 - (D) 2^4
11. The harmful substances dissolved or floating in sewage are called:
- (A) minerals we drink
 - (B) perfumes
 - (C) nutrients only
 - (D) contaminants
12. For a 30° angle, the angle that completes it to a right angle (its complement) is:
- (A) 330°
 - (B) 150°
 - (C) 60°
 - (D) 70°
13. Warm air rises above cool air because, for the same volume, warm air is:
- (A) lighter than cool air
 - (B) heavier than cool air
 - (C) exactly as heavy as cool air
 - (D) made of water only
14. Worked out fully as a repeated product, the power 2^4 has the value:
- (A) 16
 - (B) 32
 - (C) 10
 - (D) 25
15. The network of large underground pipes that carries sewage away from a town is the:
- (A) sewerage (sewer system)
 - (B) pipeline for drinking water
 - (C) canopy
 - (D) circuit

16. For a 110° angle, the angle that completes it to a straight line (its supplement) is:
- (A) 90°
 - (B) 70°
 - (C) 250°
 - (D) 20°
17. Air always moves from a region of higher pressure toward a region of:
- (A) no air at all
 - (B) equal pressure
 - (C) lower pressure
 - (D) higher pressure
18. The value of 10^3 is:
- (A) 1000
 - (B) 100
 - (C) 300
 - (D) 30
19. Sewage is cleaned before it is released back into a river at a:
- (A) weather station
 - (B) fruit-juice factory
 - (C) sewage (wastewater) treatment plant
 - (D) power plant
20. Sharing one vertex and one common arm while sitting on opposite sides of that arm, two such angles are called:
- (A) complementary angles
 - (B) alternate angles
 - (C) vertically opposite angles
 - (D) adjacent angles
21. Compared with slow or still air, faster-moving air exerts ____ pressure.
- (A) exactly the same
 - (B) higher
 - (C) no
 - (D) lower
22. Any non-zero number raised to the power 0 equals:
- (A) the number itself
 - (B) 10
 - (C) 0
 - (D) 1

23. At the treatment plant, sewage is first passed through a bar screen to remove:
- (A) the smell only
 - (B) dissolved salts
 - (C) large floating objects like rags and sticks
 - (D) germs and bacteria
24. When two straight lines cross, the two angles directly opposite each other at the crossing point are:
- (A) always different
 - (B) always 90° each
 - (C) equal (vertically opposite)
 - (D) supplementary to each other
25. When you blow hard across the top of a paper strip held to your lips, the strip:
- (A) presses flat down
 - (B) tears apart
 - (C) rises up
 - (D) stays exactly still
26. Using the rule $a^m \times a^n = a^{m+n}$, the product $3^2 \times 3^3$ equals:
- (A) 3^6
 - (B) 3^5
 - (C) 9^5
 - (D) 3^2
27. After the bar screen, the water is allowed to flow slowly so that heavy:
- (A) sand and grit settle to the bottom
 - (B) germs multiply faster
 - (C) rags float to the top
 - (D) oil rises and is bottled
28. Two adjacent angles formed on one side of a straight line add up to:
- (A) 90°
 - (B) 45°
 - (C) 360°
 - (D) 180°
29. During the day, land heats up faster than the sea, so near the coast the breeze blows from:
- (A) sky to ground
 - (B) land to sea
 - (C) sea to land
 - (D) north pole to coast

30. Using $a^m / a^n = a^{m-n}$, the quotient $5^{11} / 5^9$ equals:

- (A) 5^{11}
- (B) 5^3
- (C) 1^3
- (D) 5^2

31. In the settling tank, the solids that sink to the bottom form a thick mass called:

- (A) clear water
- (B) sludge
- (C) biogas
- (D) chlorine

32. If a linear pair has one angle of 125° , the other angle is:

- (A) 65°
- (B) 35°
- (C) 55°
- (D) 235°

33. At night the land cools faster than the sea, so near the coast the breeze blows from:

- (A) sea to land
- (B) land to sea
- (C) from the moon
- (D) straight upward only

34. Using $(a^m)^n = a^{m \cdot n}$, the value of $(2^3)^2$ equals:

- (A) 4^3
- (B) 2^6
- (C) 2^5
- (D) 2^4

35. The clearer water from the top of the settling tank is sent on, and air is bubbled through it to grow helpful:

- (A) salt crystals
- (B) fish
- (C) algae we eat
- (D) aerobic bacteria

36. Two lines in the same plane that never meet, no matter how far they are extended, are:

- (A) perpendicular lines
- (B) intersecting lines
- (C) parallel lines
- (D) curved lines

37. The huge seasonal winds that bring most of India's rain are called the:
- (A) polar winds
 - (B) land breezes
 - (C) trade-free winds
 - (D) monsoon winds
38. The value of (-1) raised to any even power is:
- (A) 0
 - (B) 2
 - (C) 1
 - (D) -1
39. Air is bubbled into the tank mainly to help the bacteria that need:
- (A) salt to grow
 - (B) oxygen to break down waste
 - (C) plastic to feed on
 - (D) darkness to hide
40. Two lines that cross to make a right angle (90°) are said to be:
- (A) curved
 - (B) parallel
 - (C) perpendicular
 - (D) vertically opposite
41. A cyclone forms over warm seas; its energy comes from the heat released when water vapour:
- (A) freezes into hailstones
 - (B) evaporates from the leaves
 - (C) condenses into droplets
 - (D) splits into atoms
42. The value of (-1) raised to any odd power is:
- (A) -2
 - (B) 0
 - (C) 1
 - (D) -1
43. The aerobic bacteria use oxygen to feed on and break down the:
- (A) human waste and other dirt in the water
 - (B) metal pipes
 - (C) the concrete tank
 - (D) the clean treated water
44. A line that cuts across two or more other lines is called a:
- (A) parallel
 - (B) radius
 - (C) transversal
 - (D) diagonal of a circle

45. The calm, almost windless centre of a cyclone is called the:

- (A) wall
- (B) root
- (C) eye
- (D) tail

46. Which is greater, 2^4 or 4^2 ?

- (A) 4^2 is greater
- (B) they are equal
- (C) neither can be compared
- (D) 2^4 is greater

47. Once settled and dried, the sludge can be put to good use as:

- (A) window glass
- (B) drinking water
- (C) cooking oil
- (D) manure for fields

48. Where a transversal cuts a pair of parallel lines, every set of corresponding angles turns out to be:

- (A) always 90°
- (B) complementary
- (C) equal
- (D) supplementary

49. Around the calm centre, the ring carrying the strongest winds and heaviest rain is the:

- (A) eye wall
- (B) eye
- (C) outer calm
- (D) horizon

50. Written using exponents, the prime factorisation of 72 is:

- (A) $2^3 \times 3^3$
- (B) $2^2 \times 3^3$
- (C) $2^3 \times 3^2$
- (D) $2^4 \times 3$

51. Sludge left to decompose without air produces a useful gas called:

- (A) oxygen
- (B) chlorine
- (C) biogas
- (D) helium

52. When a transversal crosses two parallel lines, a pair of alternate interior angles is:
- (A) unequal
 - (B) always 100°
 - (C) equal
 - (D) complementary
53. At the very centre of a cyclone the air pressure is:
- (A) very low
 - (B) very high
 - (C) zero everywhere
 - (D) exactly average
54. The number 8 written as a power of 2 is:
- (A) 3^2
 - (B) 2^4
 - (C) 2^3
 - (D) 2^2
55. Before the treated water is released into a river, it is often disinfected with:
- (A) sugar
 - (B) petrol
 - (C) sand
 - (D) chlorine
56. When a transversal crosses two parallel lines, the two co-interior (same-side interior) angles add up to:
- (A) 360°
 - (B) 180°
 - (C) 90°
 - (D) 270°
57. A dark, funnel-shaped cloud reaching from the sky to the ground with violently spinning air is a:
- (A) tornado
 - (B) sea breeze
 - (C) monsoon
 - (D) rainbow
58. In standard (scientific) form, the number 47000 is written as:
- (A) 47×10^3
 - (B) 4.7×10^4
 - (C) 4.7×10^3
 - (D) 0.47×10^5

59. Pouring cooking oil and fat down the kitchen drain is harmful mainly because it:
- (A) cleans the pipes
 - (B) blocks the drains and pipes
 - (C) makes the water drinkable
 - (D) turns into biogas instantly
60. A quarter-turn angle, measuring exactly 90° , is known as a:
- (A) straight angle
 - (B) obtuse angle
 - (C) acute angle
 - (D) right angle
61. Wind speed at a weather station is measured using an instrument called the:
- (A) rain gauge
 - (B) anemometer
 - (C) thermometer
 - (D) barometer
62. Written as an ordinary number, 6.3×10^2 is:
- (A) 6300
 - (B) 6.300
 - (C) 63
 - (D) 630
63. Used tea leaves, cotton and other solids should go in the dustbin, not the drain, because they:
- (A) choke the drains
 - (B) purify the water
 - (C) feed the river fish
 - (D) make chlorine
64. An angle that is less than 90° is called:
- (A) reflex
 - (B) obtuse
 - (C) straight
 - (D) acute
65. In different parts of the world a cyclone is also called a hurricane or a:
- (A) tsunami
 - (B) monsoon
 - (C) avalanche
 - (D) typhoon
66. A cyclone's wind blows at about 2×10^2 km/h. In ordinary numbers that speed is:
- (A) 200 km/h
 - (B) 2000 km/h
 - (C) 22 km/h
 - (D) 20 km/h

67. Open drains and stagnant pools of sewage are dangerous because they breed:

- (A) clean air
- (B) useful honeybees
- (C) fresh drinking water
- (D) disease-causing flies and mosquitoes

68. An angle between 90° and 180° is called:

- (A) acute
- (B) obtuse
- (C) right
- (D) reflex

69. As a cyclone reaches the coast it can push sea water far inland in a sudden rise called a storm:

- (A) surge
- (B) mirage
- (C) breeze
- (D) drizzle

70. One million (1 followed by six zeros) written as a power of ten is:

- (A) 6^1 ■
- (B) 10^6 ■
- (C) 10^7 ■
- (D) 10^8 ■

71. A simple, low-cost on-site pit that treats waste where there are no sewer pipes is a:

- (A) septic tank
- (B) wind tunnel
- (C) water cooler
- (D) transformer

72. An angle of exactly 180° , forming a straight line, is a:

- (A) reflex angle
- (B) straight angle
- (C) complete angle
- (D) right angle

73. During a cyclone the roof of a weak house can be lifted off because, outside, the fast wind makes the pressure:

- (A) exactly equal inside and out
- (B) lower than inside
- (C) disappear completely
- (D) higher than inside

74. Multiplying four 3's together, the power 3^4 comes to:

- (A) 81
- (B) 27
- (C) 64
- (D) 12

75. A toilet that turns human waste into manure using worms, with no sewer connection, is a:

- (A) rainwater tank
- (B) biogas cylinder
- (C) flush-only city toilet
- (D) vermi-processing (composting) toilet

76. An angle greater than 180° but less than 360° is a:

- (A) reflex angle
- (B) right angle
- (C) obtuse angle
- (D) acute angle

77. A sealed tin can with a little boiling water inside is cooled; it gets crushed because the pressure inside becomes ____ the outside air pressure.

- (A) lower than
- (B) doubled compared to
- (C) higher than
- (D) equal to

78. Using laws of exponents, $2^3 \times 5^3$ can be written as a single power:

- (A) $(2 \times 5)^3$
- (B) 7^3
- (C) 10^3
- (D) $2^3 \times 5^3$

79. Keeping ourselves and our surroundings clean and disposing of waste safely is together called:

- (A) germination
- (B) transportation
- (C) irrigation
- (D) sanitation

80. Two complementary angles are equal to each other; each one therefore measures:

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

81. The science of studying weather, clouds and storms to give warnings is done by the ____ department.

- (A) meteorological
- (B) electrical
- (C) agricultural
- (D) geological

82. The phrase 'four to the power three' means:

- (A) 4×3
- (B) $4 \times 4 \times 4$
- (C) $4 + 4 + 4$
- (D) $3 \times 3 \times 3 \times 3$

83. Poor sanitation and untreated sewage spread water-borne illnesses such as:

- (A) broken bones
- (B) short-sightedness
- (C) colour blindness
- (D) cholera and typhoid

84. Two supplementary angles are equal to each other; each one therefore measures:

- (A) 180°
- (B) 60°
- (C) 90°
- (D) 45°

85. The safest action when a cyclone warning is given is to:

- (A) move to a safe shelter on higher ground
- (B) go to the beach to watch
- (C) touch fallen electric wires
- (D) stand under a tall tree

86. Using the division rule, $10^{\blacksquare} / 10^2$ equals:

- (A) $10^{\blacksquare}$
- (B) 10^2
- (C) 1^2
- (D) $10^{\blacksquare}$

87. A worker who must enter a sewer or manhole should always:

- (A) remove all protective gear
- (B) go in alone without telling anyone
- (C) wear safety gear and check for poisonous gases first
- (D) carry an open flame inside

88. If two angles are complementary and one of them is 25° , the other is:
- (A) 155°
 - (B) 65°
 - (C) 35°
 - (D) 75°
89. Near the equator the Sun's heat is strongest, so the air there is constantly being:
- (A) frozen and kept still
 - (B) warmed and made to rise
 - (C) cooled and made to sink
 - (D) turned into rock
90. The value of 1 raised to the power 100 is:
- (A) 1
 - (B) 100
 - (C) 10
 - (D) 0
91. Treated water should be released into a river only after the harmful germs in it are:
- (A) multiplied
 - (B) destroyed
 - (C) frozen
 - (D) bottled for sale
92. If two angles are supplementary and one of them is 90° , the other is:
- (A) 90°
 - (B) 0°
 - (C) 180°
 - (D) 45°
93. Cyclones cause the worst damage along the ____ rather than deep inland.
- (A) mountain peaks
 - (B) underground caves
 - (C) desert centre
 - (D) coastline
94. A treatment tank cleans 10^3 litres of water each hour. In 10 hours, written as a power of ten, it cleans ____ litres.
- (A) 10^1
 - (B) 10^2
 - (C) 10^3
 - (D) 10^{13}

95. Pouring untreated factory chemicals into a drain is a problem because they:

- (A) turn into clean rain
- (B) speed up safe cleaning
- (C) make the water sweeter
- (D) poison the water and the bacteria that clean it

96. A weather vane points 40° east of north; to point exactly east (90° from north) it must turn a further:

- (A) 60°
- (B) 40°
- (C) 50°
- (D) 130°

97. Two light balloons hang side by side with a small gap; when you blow air through the gap, they swing:

- (A) they pop instantly
- (B) toward each other
- (C) away from each other
- (D) straight up to the ceiling

98. Which of these has the greatest value: $2^{\blacksquare}$, 4^2 , 3^2 , or 5?

- (A) 3^2
- (B) 4^2
- (C) 5
- (D) $2^{\blacksquare}$

99. If everyone reduces the waste they pour down drains, the load on a treatment plant becomes:

- (A) frozen solid
- (B) smaller and easier to clean
- (C) larger and dirtier
- (D) exactly the same forever

100. Two roads meet so that one angle between them is 120° ; the angle on the other side, along the straight road, is:

- (A) 240°
- (B) 60°
- (C) 120°
- (D) 30°